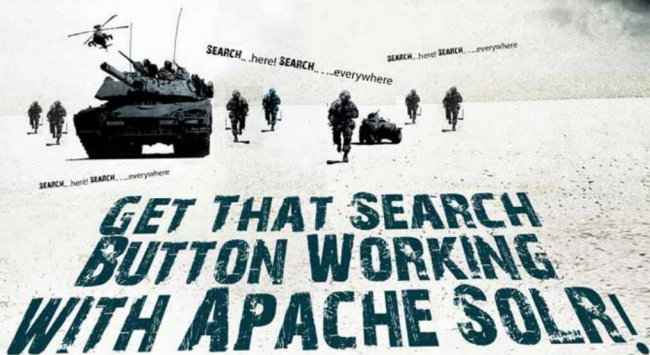




Search is probably the most important component of an application when it comes to finding valuable data, and that is precisely what Solr has been made for. Let this article guide you with setting up your own search server that will handle all the search needs of your Web application, whichever language it happens to be built in — Java, Python, PHP or anything else!

Solr is an open source search server that was originally developed at CNet, and then donated to Apache. You should only be thinking about Solr if you want to implement some kind of search facility for your Web application that's backed by random sources of data. Although it has been built with enterprise applications in mind, where users would set up instances of a search server on a separate machine dedicated for this purpose, this shouldn't stop you from using it for small projects too.

Solr is based on the full-text search library called Apache Lucene. It is mostly built as an enhancement on top of an index provided by it. Solr has been built completely in Java and so, you'll need to have Java and a servlet container like Tomcat or Glassfish installed, on top of which Solr can be deployed as a war package. However, that does not mean that you need to know Java, unless you want to make modifications to Solr!

Solr provides XML/HTTP interfaces and APIs just like Web services, and so can work with almost any programming language. The extensive external configurations available are more than enough to make customisations. Apart from this, it even provides a comprehensive administration interface with a browser. Solr also gives you a good set of default settings to start with, and provides good documentation along with it. The best applications being used out there are highly customised, but they require a deep understanding of the type of data to be searched, and the queries that users are going to

make. Information can be collected directly from file-systems, databases, websites, directories, mail servers or anything else. On the whole, Solr is a good compromise for enterprise use as well as personal applications.

Why Solr?

Whenever we think about search, what comes to our minds is usually Google; after all, it has been the world leader in terms of search technology. Then why use Solr? For starters, Solr allows you to have precise control over what gets indexed, how it gets indexed and how it's retrieved from the index. In short, a powerful search solution that is totally under your control, and can change the way users perceive and interact with your application. Setting up a separate instance of a search server will help you streamline your website and help your users wade through data with an ease that they have never experienced before.

But then, why go through the trouble of setting up a separate search server, when even databases can provide such text-indexing capabilities? Why can't we just 'SELECT * FROM Content' to retrieve data? More importantly, if Solr has been implemented on top of Lucene, why can't we build our own search application with its help? Well, building a search application on top of Lucene would require a lot of work—and you'd really have to know what you're doing (in which case, you wouldn't need to read this article at all). Talking about comparisons with databases, it depends on what kind of search